Solution

Minors :

$$M_{11} = \begin{vmatrix} -5 & 6 \\ 5 & 2 \end{vmatrix} = -10 - 30 = -40$$

$$M_{12} = \begin{vmatrix} 4 & 6 \\ -3 & 2 \end{vmatrix} = 8 + 18 = 26$$

$$M_{13} = \begin{vmatrix} 4 & -5 \\ -3 & 5 \end{vmatrix} = 20 - 15 = 5$$

$$M_{21} = \begin{vmatrix} 3 & -2 \\ 5 & 2 \end{vmatrix} = 6 + 10 = 16$$

$$M_{22} = \begin{vmatrix} 1 & -2 \\ -3 & 2 \end{vmatrix} = 2 - 6 = -4$$

$$M_{23} = \begin{vmatrix} 1 & 3 \\ -3 & 5 \end{vmatrix} = 5 + 9 = 14$$

$$M_{31} = \begin{vmatrix} 3 & -2 \\ -5 & 6 \end{vmatrix} = 18 - 10 = 8$$

$$M_{32} = \begin{vmatrix} 1 & -2 \\ 4 & 6 \end{vmatrix} = 6 + 8 = 14$$

$$M_{33} = \begin{vmatrix} 1 & 3 \\ 4 & -5 \end{vmatrix} = -5 - 12 = -17$$

Cofactors :

$$A_{11} = (-1)^{1+1}(-40) = -40$$

$$A_{12} = (-1)^{1+2}(+26) = -26$$

$$A_{13} = (-1)^{1+3}(5) = 5$$

$$A_{21} = (-1)^{2+1}(16) = -16$$

$$A_{22} = (-1)^{2+2}(-4) = -4$$

$$A_{23} = (-1)^{2+3}(14) = -14$$

$$A_{31} = (-1)^{3+1}(8) = 8$$

$$A_{32} = (-1)^{3+2}(14) = -14$$

$$A_{33} = (-1)^{3+3}(-17) = -17$$

Expanding along R_1 yields

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} \ .$$

$$|A| = 1(-40) + (3)(-26) + (-2)(5) = -128 \ . \qquad \text{... (3)}$$

Expanding along C_1 yields

$$|A| = a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31} \ .$$

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